

Fig. 4: An actual-size, single-side PCB for the live home security system

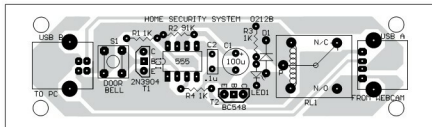


Fig. 5: Component layout for the PCB

tool with the acquired image file as an attachment. The owner's e-mail ID can be updated in this file using "xdotool type 'email_id@gmail.com';" command.

The xdotool can be installed from

the Internet by using the following command:

```
sudo apt-get install xdotool
```

This tool lets you simulate keyboard input and mouse activity, move and resize windows, etc. It uses X11's

XTEST extension and other Xlib functions. Additionally, you can modify window properties. You can use xdotool to switch desktops, move windows between desktops, and change the number of desktops.

Two commands of Xdotool are explained below:

Xdotool type "ABE"-Types ABE text

Xdotool key tab-Gives tab keystroke

With 'xdotool -h' executed in the console, the details of xdotool command can be viewed.

Mail_audio.sh. This Shell script sends an e-mail to another e-mail ID using Thunderbird tool with the recorded audio file as attachment.

An actual-size, single-side PCB layout for the live home security system is shown in Fig. 4 and its component layout in Fig. 5.

EFY note. The source codes of this article are available on www.efymag.com website.

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SOURCE CODES

```
loop.pl ->
#!/usr/bin/perl
use strict;
use warnings;
my $i=0;
while($i == 0)
{
    system "lsusb [sudo tee usblst.txt";
    open FILE, "usblst.txt" or die $!;
    my $line;
    my $count;
    my $count2=0;
    open FILE, "usblst.txt" or die $!;
    while ($line = <FILE>)
    {
        my @word = split(/ /, $line);
        $count =
        grep(/^Logitech/, @word);
        if($count==1)
        {
            $count2=1;
        }
    }
    close FILE;
    if ($count2 ==1)
    {
        my $date = localtime
        //, $date);
        my @date2 = split(/
        $date = "$date2[4]_d
        ate2[0]_date2[1]_date2[3]_date2[5]";
        system "echo $date |
        sudo tee date.txt";
        system "./webcam.pl";
        system "./listen.pl";
        system "./record.pl";
        sleep 10;
        system "./mail_pic.
        sh";
        sleep 20;
    }
}
```

```
sh";
}
}
}
}
}
```

```
webcam.pl ->
#!/usr/bin/perl
use strict;
use warnings;
open FILE, "date.txt" or die $!;
my $date = <FILE>;
chop($date);
print "\nhi\n$date";
system ("sudo streamer -c /dev/video0 -b
16 -o /home/efyindia/Documents/photos/
image_$date.jpeg");
close FILE;
```

```
listen.pl ->
#!/usr/bin/perl
use strict;
use warnings;
system ("sudo mplayer attention.wav");
```

```
record.pl ->
#!/usr/bin/perl
use strict;
use warnings;
open FILE, "date.txt" or die $!;
my $date = <FILE>;
chop($date);
system ("sudo arecord -f CD -d 5 -D
hw:0,0 /home/efyindia/Documents/audios/
audio_$date.wav");
```

```
mail_pic.sh ->
#!/bin/bash -x
while read myline
do
    echo $myline
    date=$myline
```

```
done < date.txt
thunderbird -compose "attachment=file:///
home/efyindia/Documents/photos/
image_$date.jpeg"s
sleep 5;
xdotool type 'email_id@gmail.com';
sleep 1;
xdotool key Tab;
sleep 1;
xdotool type 'somebody is at the door';
sleep 1;
xdotool key Tab;
sleep 1;
xdotool type 'this is the picture';
sleep 1;
xdotool key Ctrl+Return;
```

```
mail_audio.sh ->
#!/bin/bash -x
while read myline
do
    echo $myline
    date=$myline
done < date.txt
thunderbird -compose "attachment=file:///
home/efyindia/Documents/audios/
audio_$date.wav"s
sleep 5;
xdotool type 'email_id@gmail.com';
sleep 1;
xdotool key Tab;
sleep 1;
xdotool type 'somebody is at the door';
sleep 1;
xdotool key Tab;
sleep 1;
xdotool type 'this is the audio';
sleep 1;
xdotool key Ctrl+Return;
```